Hellenic Complex Systems Laboratory

Calculation of Diagnostic Accuracy Measures

Technical Report XVII

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Calculation of Diagnostic Accuracy Measures

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Short Description of the Demonstration

This Demonstration shows calculations of point estimations and confidence intervals for various accuracy measures of a diagnostic test for a disease. This is done for differing negative and positive test results of nondiseased and diseased populations and differing confidence levels 1-α for the estimations of the lower and upper confidence limits. The calculated measures are the sensitivity (*Se*), the specificity (*Sp*), the positive predictive value (*PPV*), the negative predictive value (*NPV*), the (diagnostic) odds ratio (*OR*), the likelihood ratio for a positive test result (LR^+) and the likelihood ratio for a negative test result (LR^-). The measures can be selected using the menu. The negative and positive test results of the nondiseased and diseased populations, along with the confidence level, are chosen using the sliders.

Search Terms

sensitivity, specificity, diagnostic test, clinical accuracy, diagnostic accuracy, normal distribution, positive predictive value, negative predictive value, likelihood ratio, odds ratio, confidence level, confidence intervals

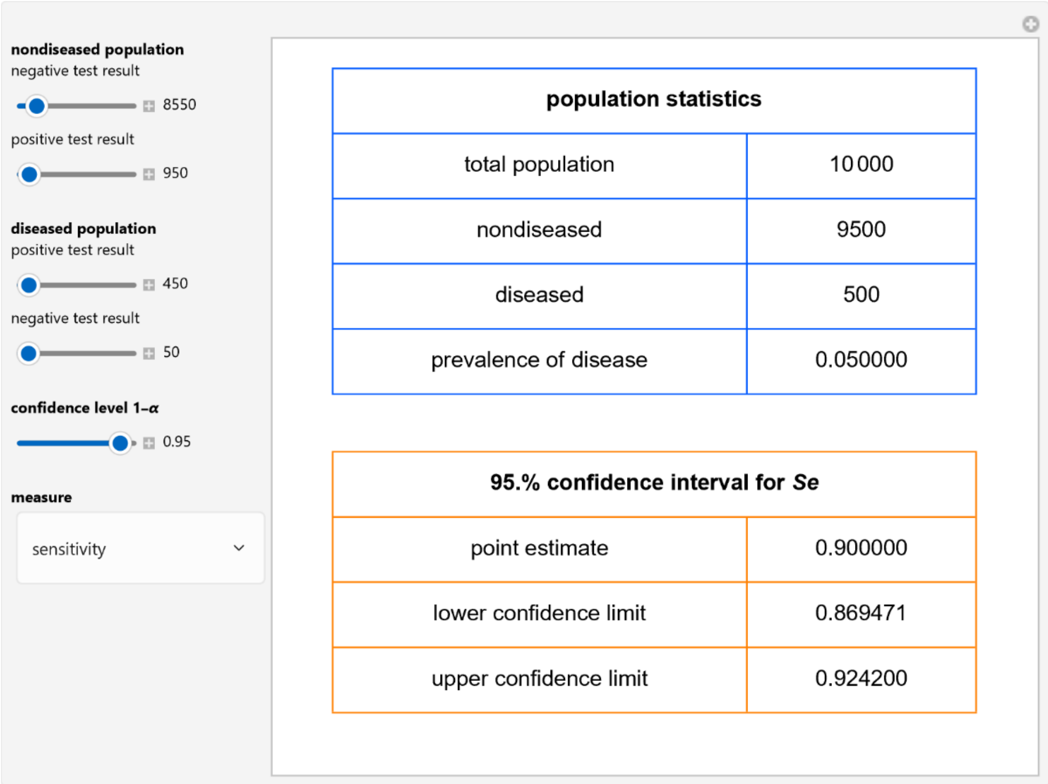

Figure 1: *Population statistics and the point estimation and the 95.0% confidence interval for the sensitivity of a diagnostic test, with the settings shown at the left.*

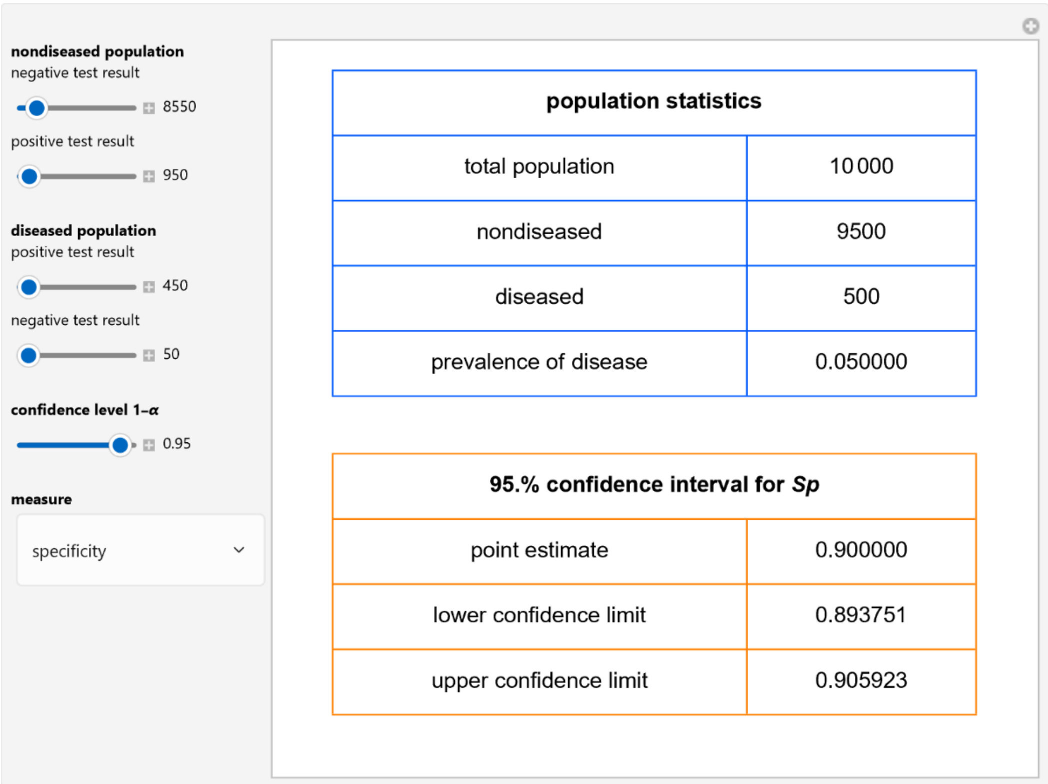

Figure 2: *Population statistics and the point estimation and the 95.0% confidence interval for the specificity of a diagnostic test, with the settings shown at the left.*

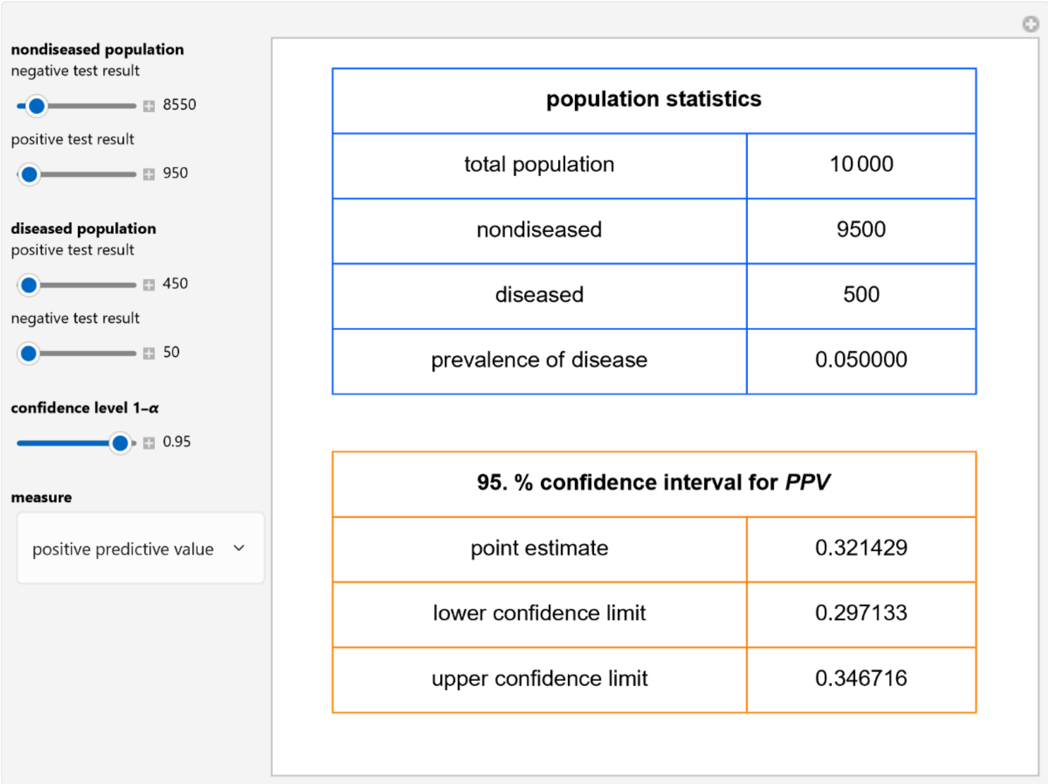

Figure 3: *Population statistics and the point estimation and the 95.0% confidence interval for the positive predictive value of a diagnostic test, with the settings shown at the left.*

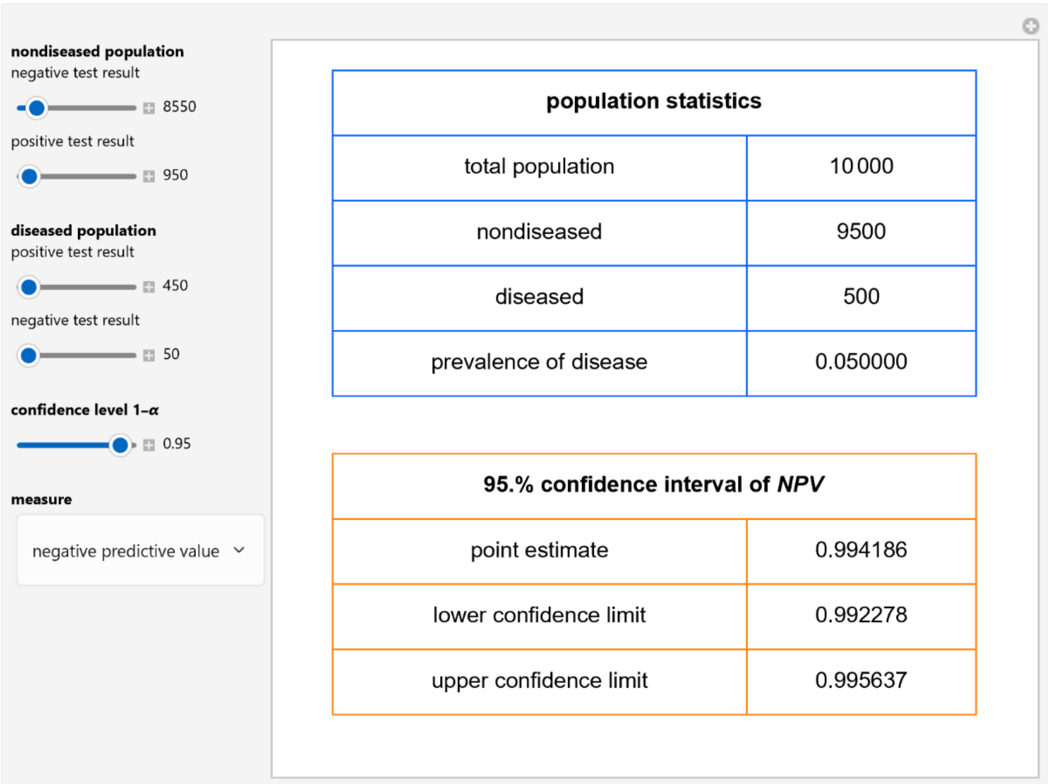

Figure 4: *Population statistics and the point estimation and the 95.0% confidence interval for the negative predictive value of a diagnostic test, with the settings shown at the left.*

Details

Sensitivity refers to the fraction of the diseased population with a positive test result, while specificity is the fraction of the nondiseased population with a negative test result. Positive predictive value is the fraction of the population with a positive test result that is diseased, while negative predictive value is the fraction of the population with a negative test result that is not diseased. Prevalence of the disease is the ratio of the diseased population to the total (nondiseased and diseased) population. If we denote by *sens* the sensitivity, *Sp* the specificity, and v the prevalence, we have:

$$PPV = \frac{Se\ v}{Se\ v + (1 - Sp)(1 - v)}$$

$$NPV = \frac{Sp\ (1 - v)}{Sp\ (1 - v) + (1 - Se)v}$$

$$OR = \frac{\frac{Se}{1 - Se}}{\frac{1 - Sp}{Sp}}$$

$$LR^{+} = \frac{Se}{1 - Sp}$$

$$LR^{-} = \frac{1 - Se}{Sp}$$

The Wilson score method with continuity correction is used [1] for calculating the confidence intervals of the sensitivity, the specificity, the positive predictive value and the negative predictive value.

For the calculation of the confidence intervals of the (diagnostic) odds ratio, the likelihood ratio for a positive test result and the likelihood ratio for a negative test result, it is assumed that their natural logarithms have asymptotically normal distributions [1].

This Demonstration is an extension of another Demonstration [2] and is appropriate as an educational and research tool.

References

[1] J. L. Fleiss, B. Levin and M. C. Paik. Statistical Methods for Rates and Proportions*,* 3rd ed. *J. Wiley.* 2003.

[2] Chatzimichail T. Calculator for Diagnostic Accuracy Measures. *Wolfram Demonstrations Project*. Wolfram Research; 2018. Available at: https://www.hcsl.com/Tools/Demonstrations/CalculatorForDiagnosticAccuracyMeasures.nb

Demonstration

DOI

Source

Programming language: Wolfram Language
Availability: The updated source notebook is available at:
https://www.hcsl.com/Tools/Demonstrations/CalculationOfDiagnosticAccuracyMeasures.nb

First published in 2018 as part of the Wolfram Demonstrations Project:

https://demonstrations.wolfram.com/CalculationOfDiagnosticAccuracyMeasures/

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: https://www.wolfram.com/player/ or Wolfram Mathematica®.

System Requirements
Processor: x86-64 compatible CPU.
System memory (RAM): 4 GB+ recommended.

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